

Question 1 — IoT Sensor Dashboard

Problem 1 — Sensor Reading Classifier

```
//=====
=====
// Name      : AssQue1.cpp
// Author    :
// Version    :
// Copyright  : Your copyright notice
// Description : Hello World in C++, Ansi-style
//=====
=====
```

```
#include <iostream>
using namespace std;
```

```
int main() {
    int statuscode;
    double c;

    cout << "Enter the value of c : "<<endl;
    cin >> c;

    double fahrenheit = (c * 9 / 5) + 32;
    cout <<fahrenheit;
    cout <<endl;

    if(c < 0)
    {
        statuscode = -1;
    }
    else if(c >= 0 && c <= 29)
    {
        statuscode = 0;
    }
    else if(c >= 30 && c <= 44)
    {
        statuscode = 1;
    }

    }else if(c >= 45 && c <= 59)
    {
        statuscode = 2;
    }
}
```

```

else
{
    statuscode = 3;
}

switch(statuscode)
{
case -1 :
    cout << "SENSOR_ERROR" <<endl;
    cout << "Sensor fault — check wiring" <<endl;
    break;

case 0 :

    cout << "NORMAL" <<endl;
    cout << "No action required" <<endl;
    break;

case 1 :

    cout << "WARNING" <<endl;
    cout << "Alert sent to supervisor" <<endl;
    break;

case 2 :
    cout << "CRITICAL" <<endl;
    cout << "Cooling system triggered" <<endl;
    break;

case 3 :
    cout << "SHUTDOWN" <<endl;
    cout << "Emergency shutdown initiated" <<endl;
    break;
}

cout << "reading : " << (c >= 25 ? "Above Average" : "Below Average");

return 0;
}

```

```
Problems Tasks Console X Include Browser Terminal Eclipse IDE for C/C++ Developers 2026-09 RC1
<terminated> (exit value: 0) AssQue1.exe [C/C++ Application] D:\CDAC\CPP\AssQue1\Debug\AssQue1.exe (8/30/26, 11:32 AM)
Enter the value of c :
47.3
117.14
CRITICAL
Cooling system triggered
reading : Above Average
```

Problem 2 — Sensor Log Buffer

```
//=====
=====
// Name      : 2.cpp
// Author    :
// Version   :
// Copyright  : Your copyright notice
// Description : Hello World in C++, Ansi-style
//=====
=====

#include <iostream>
#include <iomanip>
using namespace std;

int main()
{
    int n;

    // Array can store maximum 100 readings
    float temp[100];

    // Take number of readings
    cout << "Readings entered : ";
    cin >> n;

    // Take N temperature values
    for(int i = 0; i < n; i++)
    {
        cin >> temp[i];
    }
}
```

```
}
```

```
// Variables for analysis
```

```
int errors = 0;
```

```
int normal = 0;
```

```
int warning = 0;
```

```
int critical = 0;
```

```
int shutdown = 0;
```

```
float min = 9999;
```

```
float max = -9999;
```

```
float sum = 0;
```

```
int valid = 0;
```

```
cout << "Valid readings  : ";
```

```
for(int i = 0; i < n; i++)
```

```
{
```

```
    if(temp[i] < 0)
```

```
    {
```

```
        errors++;
```

```
        continue;
```

```
    }
```

```
    cout << fixed << setprecision(1)
```

```
        << temp[i] << " ";
```

```
    if(temp[i] < min)
```

```
    {
```

```
        min = temp[i];
```

```
    }
```

```
    if(temp[i] > max)
```

```
    {
```

```
        max = temp[i];
```

```
    }
```

```

sum = sum + temp[i];

valid++;

if(temp[i] >= 20 && temp[i] < 30)
{
    normal++;
}

else if(temp[i] < 45)
{
    warning++;
}

else if(temp[i] < 60)
{
    critical++;
}

else
{
    shutdown++;
}
}

for(int i = 0; i < n; i++)
{
    if(temp[i] >= 45)
    {
        cout << "\nFirst CRITICAL : Index "
            << i << " -> " << temp[i] << " C";

        break;
    }
}

cout << "\nSkipped (errors) : " << errors;
cout << fixed << setprecision(2);

cout << "\nMin : " << min << " C";

cout << "    Max : " << max << " C";

```

```

    cout << "   Avg : " << sum / valid << " C";

    cout << "\nNormal:" << normal;

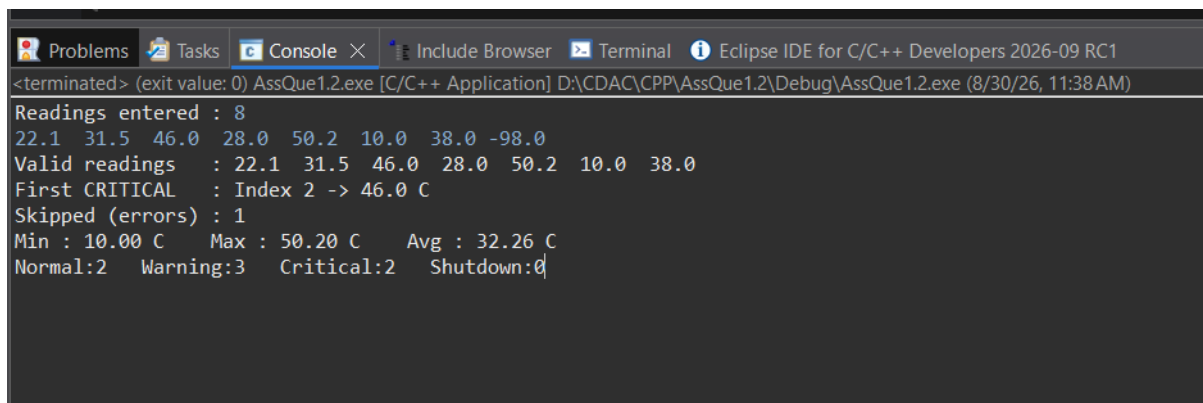
    cout << "   Warning:" << warning;

    cout << "   Critical:" << critical;

    cout << "   Shutdown:" << shutdown;

    return 0;
}

```



The screenshot shows the Eclipse IDE console window. The title bar indicates the application is 'AssQue1.2.exe [C/C++ Application]' running at 'D:\CDAC\CPP\AssQue1.2\Debug\AssQue1.2.exe (8/30/26, 11:38 AM)'. The console output displays the following information:

```

Readings entered : 8
22.1 31.5 46.0 28.0 50.2 10.0 38.0 -98.0
Valid readings   : 22.1 31.5 46.0 28.0 50.2 10.0 38.0
First CRITICAL  : Index 2 -> 46.0 C
Skipped (errors) : 1
Min : 10.00 C    Max : 50.20 C    Avg : 32.26 C
Normal:2   Warning:3   Critical:2   Shutdown:0

```

Problem 3 — Building Sensor Grid

```

//=====

=====

// Name      : 3.cpp
// Author    :
// Version   :
// Copyright  : Your copyright notice
// Description : Hello World in C++, Ansi-style
//=====

=====

#include <iostream>
#include <iomanip>
using namespace std;

int main() {

```

```

float temp[3][3];
cout << "Enter Temperatures for 3 floors and 3 rooms :\n\n";

for(int i = 0; i < 3; i++)
{
    cout << "Floor " << i + 1 << " : ";

    for(int j = 0; j < 3; j++)
    {
        cin >> temp[i][j];
    }
}

cout<<endl;
cout<<"      Room1  Room2  Room3" <<endl;

for(int i = 0; i < 3; i++)
{
    cout << "Floor" << i + 1 << " : ";

    for(int j = 0; j < 3; j++)
    {
        cout << fixed << setprecision(1) <<temp[i][j] << "  ";
    }

    cout << endl;
}

float hottest = temp[0][0];
int hotfloor = 0;
int hotroom = 0;

for(int i = 0; i < 3; i++)
{
    for(int j = 0; j < 3; j++)
    {
        if(temp[i][j] > hottest)
        {
            hottest = temp[i][j];

            hotfloor = i;
            hotroom =j;
        }
    }
}

```

```

    }
}

cout << endl;
cout<<"Hottest Room : Floor "<<hotfloor + 1 << ", Room " << hotroom + 1<< " -> "
<< hottest << " C";
float highestAvg = -9999;
int hottestFloor = 0;

for(int i = 0; i < 3; i++)
{
    float sum = 0;

    // Add temperatures of one floor
    for(int j = 0; j < 3; j++)
    {
        sum = sum + temp[i][j];
    }

    // Calculate average of that floor
    float avg = sum / 3;

    // Check if this floor has higher average
    if(avg > highestAvg)
    {
        highestAvg = avg;
        hottestFloor = i;
    }
}

cout << "\nHottest Floor : Floor " << hottestFloor + 1 << " (avg " << fixed <<
setprecision(2) << highestAvg << " C)";
int warning = 0;

for(int i = 0; i < 3; i++)
{
    for(int j = 0; j < 3; j++)
    {
        // 30 C or above = WARNING
        if(temp[i][j] >= 30)
        {
            warning++;
        }
    }
}

```



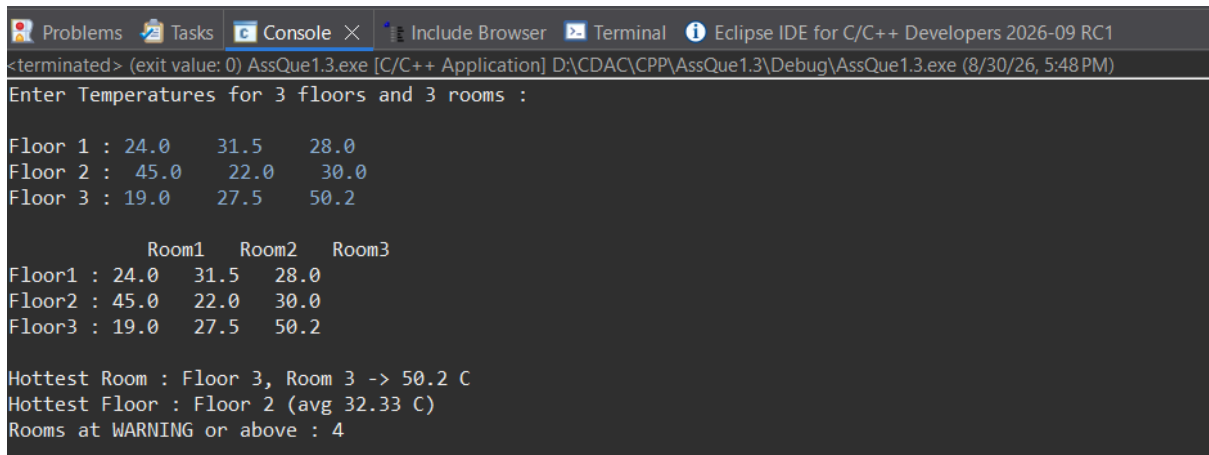
```

    }
}

cout << "\nRooms at WARNING or above : "
    << warning;

return 0;
}

```



The screenshot shows the Eclipse IDE console with the following output:

```

<terminated> (exit value: 0) AssQue1.3.exe [C/C++ Application] D:\CDAC\CPP\AssQue1.3\Debug\AssQue1.3.exe (8/30/26, 5:48 PM)
Enter Temperatures for 3 floors and 3 rooms :

Floor 1 : 24.0    31.5    28.0
Floor 2 : 45.0    22.0    30.0
Floor 3 : 19.0    27.5    50.2

      Room1    Room2    Room3
Floor1 : 24.0    31.5    28.0
Floor2 : 45.0    22.0    30.0
Floor3 : 19.0    27.5    50.2

Hottest Room : Floor 3, Room 3 -> 50.2 C
Hottest Floor : Floor 2 (avg 32.33 C)
Rooms at WARNING or above : 4

```

Problem 4 — Startup Configuration via CLI

```

//=====

=====

// Name      : 4.cpp
// Author    :
// Version   :
// Copyright  : Your copyright notice
// Description : Hello World in C++, Ansi-style

//=====

=====

#include <iostream>

#include <cstdlib>

using namespace std;

int main(int argc, char *argv[])

```

```

{
    // Check if arguments are missing
    if(argc != 4)
    {
        cout << "Usage  : ./sensor_monitor" << endl;
        cout << "Error  : Missing arguments." << endl;
        return 1;
    }

    int warn = atoi(argv[1]);
    int critical = atoi(argv[2]);
    int num = atoi(argv[3]);

    // Check warning and critical values
    if(warn >= critical)
    {
        cout << "Error  : warn must be less than critical." << endl;
        return 1;
    }

    // Check number of readings
    if(num < 1 || num > 500)
    {
        cout << "Error  : num_readings must be between 1 and 500." << endl;
        return 1;
    }

    int normal = 0;
    int warning = 0;

```

```

int criticalCount = 0;

int shutdown = 0;


// Generate readings and classify them
for(int i = 0; i < num; i++)
{
    int temp = rand() % 70;


    if(temp < warn)
        normal++;
    else if(temp < critical)
        warning++;
    else if(temp < 60)
        criticalCount++;
    else
        shutdown++;
}


cout << "Config : Warn=" << warn
    << "°C Critical=" << critical
    << "°C Readings=" << num << endl;


cout << "Results : Normal:" << normal
    << " Warning:" << warning
    << " Critical:" << criticalCount
    << " Shutdown:" << shutdown << endl;


return 0;
}

```

```
Problems Tasks Console X Include Browser Terminal Eclipse IDE for C/C++ Developers 2026-09 RC1
<terminated> (exit value: 0) AssQue1.4.exe [C/C++ Application] D:\CDAC\CPP\AssQue1.4\Debug\AssQue1.4.exe (8/29/26, 10:06 PM)
Config : Warn=30°C Critical=45°C Readings=10
Results : Normal:2 Warning:5 Critical:2 Shutdown:1
```

Question 2 — Embedded Memory Manager & Signal Processor

Problem 1 — The Bug in Sensor Recalibration

```
//=====
=====
// Name      : 1.cpp
// Author    :
// Version   :
// Copyright  : Your copyright notice
// Description : Hello World in C++, Ansi-style
//=====
=====
```

```
#include <iostream>
using namespace std;
```

```
// Call by Value
void resetSensorPairV1(int reading1, int reading2)
{
    int temp = reading1;
    reading1 = reading2;
    reading2 = temp;
}
```

```
// Call by Reference
void resetSensorPairV2(int& reading1, int& reading2)
{
    int temp = reading1;
    reading1 = reading2;
    reading2 = temp;
}
```

```
// Call by Pointer
void resetSensorPairV3(int* reading1, int* reading2)
```

```

{
    int temp = *reading1;
    *reading1 = *reading2;
    *reading2 = temp;
}

int main()
{
    int A = 55;
    int B = 12;

    // V1
    cout << "V1: Call by Value" << endl;
    cout << "Before : A=" << A << " B=" << B << endl;

    resetSensorPairV1(A, B);

    cout << "After : A=" << A << " B=" << B << endl;

    // V2
    cout << "\nV2: Call by Reference" << endl;
    cout << "Before : A=" << A << " B=" << B << endl;

    resetSensorPairV2(A, B);

    cout << "After : A=" << A << " B=" << B << endl;

    // V3
    cout << "\nV3: Call by Pointer" << endl;
    cout << "Before : A=" << A << " B=" << B << endl;

    resetSensorPairV3(&A, &B);

    cout << "After : A=" << A << " B=" << B << endl;

    return 0;
}

```

```
Problems Tasks Console X Include Browser Terminal Eclipse IDE for C/C++ Developers 2026-09 RC1
<terminated> (exit value: 0) AssQue2.1.exe [C/C++ Application] D:\CDAC\CPP\AssQue2.1\Debug\AssQue2.1.exe (8/30/26, 5:58 PM)

V1: Call by Value
Before : A=55 B=12
After  : A=55 B=12

V2: Call by Reference
Before : A=55 B=12
After  : A=12 B=55

V3: Call by Pointer
Before : A=12 B=55
After  : A=55 B=12
```

Problem 2 — Signal Processing Pipeline

```
//=====
=====
// Name      : 2.cpp
// Author    :
// Version   :
// Copyright  : Your copyright notice
// Description : Hello World in C++, Ansi-style
//=====
=====

#include <iostream>
#include <cmath>
using namespace std;

// Calculate RMS
double computeRMS(double* signal, int n)
{
    double sum = 0;

    for(int i = 0; i < n; i++)
    {
        sum = sum + (*(signal + i) * *(signal + i));
    }

    return sqrt(sum / n);
}
```

```

// Normalize the signal
void normalise(double* signal, int n)
{
    double max = 0;

    // Find maximum absolute value
    for(int i = 0; i < n; i++)
    {
        if(fabs(*(signal + i)) > max)
        {
            max = fabs(*(signal + i));
        }
    }

    // Divide every value by max
    for(int i = 0; i < n; i++)
    {
        *(signal + i) = *(signal + i) / max;
    }
}

// Count zero crossings
int countZeroCrossings(double* signal, int n)
{
    int count = 0;

    for(int i = 0; i < n - 1; i++)
    {
        // Check if two adjacent values have opposite signs
        if((*(signal + i) < 0 && *(signal + i + 1) > 0) ||
            (*(signal + i) > 0 && *(signal + i + 1) < 0))
        {
            count++;
        }
    }

    return count;
}

// Apply gain
void applyGain(double* signal, int n, double gainFactor)

```

```

{
    for(int i = 0; i < n; i++)
    {
        *(signal + i) = *(signal + i) * gainFactor;
    }
}

```

```

// Print array
void printArray(double* signal, int n)
{
    for(int i = 0; i < n; i++)
    {
        cout << *(signal + i) << " ";
    }

    cout << endl;
}

```

```

int main()
{
    double signal[] = {0.5, -1.2, 0.8, -0.3, 1.0, -0.9, 0.1};

    int n = 7;

    double gainFactor = 2.0;

    cout << "Before processing : ";
    printArray(signal, n);

    // Normalize
    normalise(signal, n);

    cout << "After normalise : ";
    printArray(signal, n);

    // Apply gain
    applyGain(signal, n, gainFactor);
}

```



```

    cout << "After applyGain  : ";
    printArray(signal, n);

    // Calculate RMS
    cout << "RMS : " << computeRMS(signal, n) << endl;

    // Count zero crossings
    cout << "Zero Crossings : "
        << countZeroCrossings(signal, n) << endl;

    return 0;
}

```

```

<terminated> (exit value: 0) AssQue2.2.exe [C/C++ Application] D:\CDAC\CPP\AssQue2.2\Debug\AssQue2.2.exe (8/30/26, 6:00 PM)
Before processing : 0.5 -1.2 0.8 -0.3 1 -0.9 0.1
After normalise   : 0.416667 -1 0.666667 -0.25 0.833333 -0.75 0.0833333
After applyGain   : 0.833333 -2 1.33333 -0.5 1.66667 -1.5 0.166667
RMS : 1.29713
Zero Crossings : 6

```

Problem 3 — Hardware Register Access

```

//=====
//=====
// Name      : 3.cpp
// Author    :
// Version   :
// Copyright  : Your copyright notice
// Description : Hello World in C++, Ansi-style
//=====
//=====

#include <iostream>
using namespace std;

int main()
{
    int statusReg = 0b10110001;
    int controlReg = 0b00000000;

```

```
int dataReg = 0b11001010;

// regPtr1: pointer to const
const int* regPtr1 = &statusReg;

cout << "Status Register : " << *regPtr1 << endl;

regPtr1 = &dataReg;

int* const regPtr2 = &controlReg;

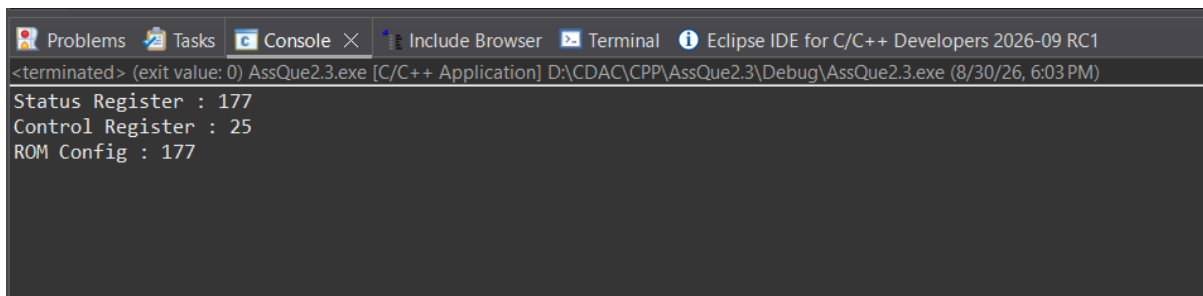
*regPtr2 = 25;

cout << "Control Register : " << *regPtr2 << endl;

const int* const regPtr3 = &statusReg;

cout << "ROM Config : " << *regPtr3 << endl;

return 0;
}
```

A screenshot of the Eclipse IDE's console window. The window has a dark theme and shows the output of a C++ program. The title bar includes tabs for 'Problems', 'Tasks', 'Console', 'Include Browser', 'Terminal', and 'Eclipse IDE for C/C++ Developers 2026-09 RC1'. The console text shows the program's execution path and the values of three registers: Status Register (177), Control Register (25), and ROM Config (177).

```
<terminated> (exit value: 0) AssQue2.3.exe [C/C++ Application] D:\CDAC\CPP\AssQue2.3\Debug\AssQue2.3.exe (8/30/26, 6:03 PM)
Status Register : 177
Control Register : 25
ROM Config : 177
```

Problem 4 — Calibration Packet Parser

```
//=====
=====

// Name      : 4.cpp
// Author    :
// Version    :
// Copyright  : Your copyright notice
// Description: Hello World in C++, Ansi-style
//=====
=====

#include <iostream>
using namespace std;

bool parsePacket(const int* rawData, int size,
                 int** outMin, int** outMax)
{
    if(size <= 0)
    {
        return false;
    }

    const int* min = rawData;
    const int* max = rawData;

    for(int i = 1; i < size; i++)
    {
        if(*(rawData + i) < *min)
        {
            min = rawData + i;
        }

        if(*(rawData + i) > *max)
        {
            max = rawData + i;
        }
    }

    *outMin = const_cast<int*>(min);
    *outMax = const_cast<int*>(max);

    return true;
}
```

```

    }

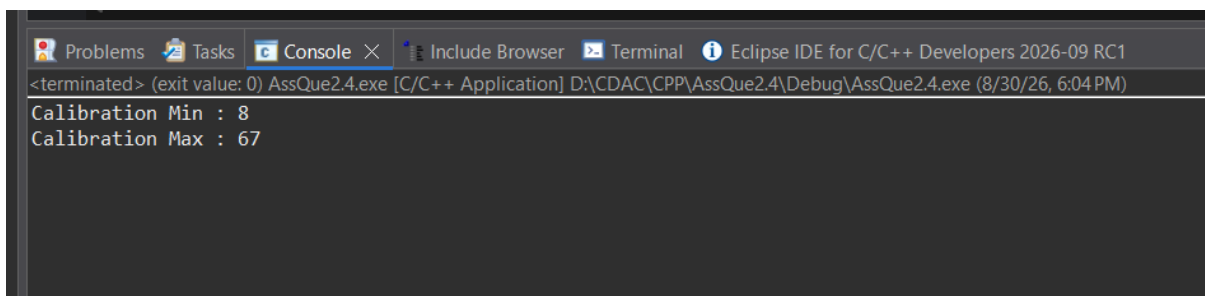
    int main()
    {
        int packet[] = {45, 12, 67, 8, 55, 31};

        int* minPtr = nullptr;
        int* maxPtr = nullptr;

        if (parsePacket(packet, 6, &minPtr, &maxPtr)) {
            cout << "Calibration Min : " << *minPtr << endl;
            cout << "Calibration Max : " << *maxPtr << endl;
        }

        return 0;
    }

```



The screenshot shows the Eclipse IDE interface with the 'Console' tab selected. The console output displays the results of the program execution: 'Calibration Min : 8' and 'Calibration Max : 67'. The status bar at the bottom indicates the application is 'AssQue2.4.exe' running on '8/30/26, 6:04 PM'.

```

<terminated> (exit value: 0) AssQue2.4.exe [C/C++ Application] D:\CDAC\CPP\AssQue2.4\Debug\AssQue2.4.exe (8/30/26, 6:04 PM)
Calibration Min : 8
Calibration Max : 67

```

Problem 5 — Drone Navigation Utilities

```

//=====

=====
// Name      : 5.cpp
// Author    :
// Version    :
// Copyright  : Your copyright notice
// Description : Hello World in C++, Ansi-style
//=====

=====

#include <iostream>
#include <cmath>
using namespace std;

// Find distance between two points
inline double distanceBetween(double x1, double y1, double x2, double y2)

```

```

{
    return sqrt(pow(x2 - x1, 2) + pow(y2 - y1, 2));
}

// Convert degrees to radians
inline double toRadians(double degrees)
{
    return degrees * (M_PI / 180.0);
}

// Restrict value between minVal and maxVal
inline double clamp(double value, double minVal, double maxVal)
{
    if(value < minVal)
        return minVal;

    if(value > maxVal)
        return maxVal;

    return value;
}

// Check whether point is inside safe zone
inline bool isInSafeZone(double x, double y,
                        double cx, double cy, double radius)
{
    return distanceBetween(x, y, cx, cy) <= radius;
}

int main()
{
    double homeX = 0.0;
    double homeY = 0.0;
    double radius = 50.0;

    // Three waypoints
    double x1 = 10.0, y1 = 20.0;
    double x2 = 30.0, y2 = 40.0;
    double x3 = 60.0, y3 = 10.0;

    // Waypoint 1
    cout << "Waypoint 1 : Distance = "
        << distanceBetween(homeX, homeY, x1, y1);

```

```

if(isInSafeZone(x1, y1, homeX, homeY, radius))
    cout << " Inside Safe Zone" << endl;
else
    cout << " Outside Safe Zone" << endl;

// Waypoint 2
cout << "Waypoint 2 : Distance = "
    << distanceBetween(homeX, homeY, x2, y2);

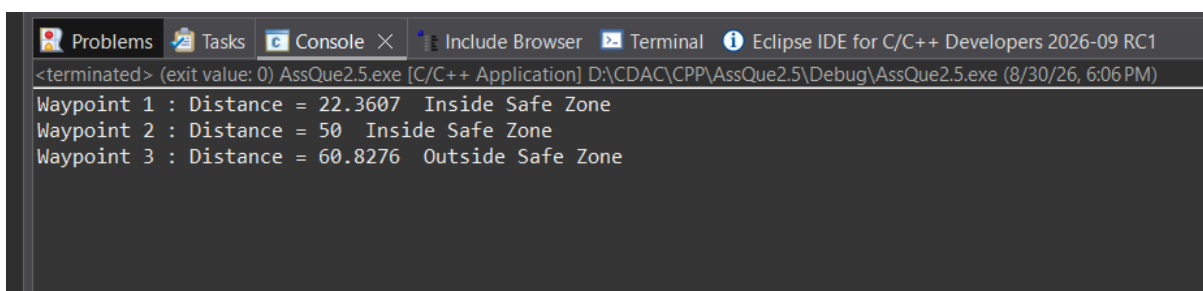
if(isInSafeZone(x2, y2, homeX, homeY, radius))
    cout << " Inside Safe Zone" << endl;
else
    cout << " Outside Safe Zone" << endl;

// Waypoint 3
cout << "Waypoint 3 : Distance = "
    << distanceBetween(homeX, homeY, x3, y3);

if(isInSafeZone(x3, y3, homeX, homeY, radius))
    cout << " Inside Safe Zone" << endl;
else
    cout << " Outside Safe Zone" << endl;

return 0;
}

```



The screenshot shows the Eclipse IDE interface with the 'Console' tab selected. The console output displays the results of the program execution for three waypoints. The first two waypoints are inside the safe zone, while the third is outside.

```

<terminated> (exit value: 0) AssQue2.5.exe [C/C++ Application] D:\CDAC\CPP\AssQue2.5\Debug\AssQue2.5.exe (8/30/26, 6:06 PM)
Waypoint 1 : Distance = 22.3607 Inside Safe Zone
Waypoint 2 : Distance = 50 Inside Safe Zone
Waypoint 3 : Distance = 60.8276 Outside Safe Zone

```

Question 3 — HR Payroll Engine

```
//=====
=====
// Name      : AssQue3.cpp
// Author    :
// Version    :
// Copyright  : Your copyright notice
// Description: Hello World in C++, Ansi-style
//=====
=====
```

```
#include <iostream>
#include <iomanip>
using namespace std;
```

```
class Employee
{
private:
    int empId;
    string name;
    string department;
    char grade;
    double basicSalary;
    bool isActive;

    static int employeeCount;
```

```
public:
```

```
    Employee()
    {
        empId = 1001 + employeeCount;
        employeeCount++;

        name = "";
        department = "";
        grade = 'D';
        basicSalary = 10001;
        isActive = true;
    }
```

```
    void setName(const string& n)
```

```

{
    if(n != "")
        name = n;
    else
        cout << "ERROR: Name cannot be empty." << endl;
}

void setDepartment(const string& d)
{
    if(d == "Engineering" || d == "HR" ||
        d == "Finance" || d == "Operations")
    {
        department = d;
    }
    else
    {
        cout << "ERROR: " << d
            << " is not a registered department." << endl;
    }
}

void setGrade(char g)
{
    if(g == 'A' || g == 'B' || g == 'C' || g == 'D')
        grade = g;
    else
        cout << "ERROR: Invalid grade " << g
            << ". Accepted values: A, B, C, D." << endl;
}

void setBasicSalary(double s)
{
    if(s > 10000 && s < 500000)
        basicSalary = s;
    else
        cout << "ERROR: Salary must be between Rs.10,000 and "
            << "Rs.5,00,000. Value rejected." << endl;
}

void deactivate()
{
    isActive = false;
}

```



```
int getEmpId() const
{
    return empId;
}

string getName() const
{
    return name;
}

string getDepartment() const
{
    return department;
}

char getGrade() const
{
    return grade;
}

double getBasicSalary() const
{
    return basicSalary;
}

bool getIsActive() const
{
    return isActive;
}

double computeAllowances() const
{
    if(grade == 'A')
        return basicSalary * 40 / 100;
    else if(grade == 'B')
        return basicSalary * 30 / 100;
    else if(grade == 'C')
        return basicSalary * 20 / 100;
    else
        return basicSalary * 10 / 100;
}
```

```

double computeGrossSalary() const
{
    return basicSalary + computeAllowances();
}

double computeTax() const
{
    double gross = computeGrossSalary();

    if(gross <= 50000)
        return 0;
    else if(gross <= 100000)
        return (gross - 50000) * 10 / 100;
    else
        return 5000 + (gross - 100000) * 20 / 100;
}

double computeNetSalary() const
{
    return computeGrossSalary() - computeTax();
}

void acceptDetails()
{
    string n, d;
    char g;
    double s;

    cout << "Enter name: ";
    cin >> ws;
    getline(cin, n);
    setName(n);

    cout << "Enter department: ";
    cin >> d;
    setDepartment(d);

    cout << "Enter grade: ";
    cin >> g;
    setGrade(g);

    cout << "Enter basic salary: ";
    cin >> s;
}

```

```

        setBasicSalary(s);
    }

void printPayslip() const
{
    cout << "\n===== " << endl;
    cout << "      EMPLOYEE PAYSIP" << endl;
    cout << "===== " << endl;

    cout << "Emp ID      : " << empId << endl;
    cout << "Name        : " << name << endl;
    cout << "Department  : " << department << endl;
    cout << "Grade       : " << grade << endl;

    if(isActive)
        cout << "Status      : Active" << endl;
    else
        cout << "Status      : Inactive" << endl;

    cout << "-----" << endl;

    cout << fixed << setprecision(2);

    cout << "Basic Salary : Rs. " << basicSalary << endl;
    cout << "Allowance   : Rs. " << computeAllowances() << endl;
    cout << "Gross Salary : Rs. " << computeGrossSalary() << endl;
    cout << "Tax         : Rs. " << computeTax() << endl;
    cout << "Net Salary  : Rs. " << computeNetSalary() << endl;

    cout << "===== " << endl;
}

static int getEmployeeCount()
{
    return employeeCount;
}

};

int Employee::employeeCount = 0;
int main()
{
    Employee e1;
    Employee *e2 = new Employee;

```

```

Employee *e3 = new Employee;
e1.acceptDetails();
e2->acceptDetails();
e3->acceptDetails();

e1.printPayslip();
e2->printPayslip();
e3->printPayslip();

e3->deactivate();

if(!e3->getIsActive())
    cout << e3->getName()
        << " is no longer active. Payroll skipped."
        << endl;

cout << "Total Employees : "
    << Employee::getEmployeeCount() << endl;

delete e2;
delete e3;

return 0;
}

```

```

=====
EMPLOYEE PAYSIP
=====
Emp ID      : 1001
Name       : Priya Sharma
Department  : Engineering
Grade      : B
Status     : Active
-----
Basic Salary : Rs. 75000.00
Allowance    : Rs. 22500.00
Gross Salary : Rs. 97500.00
Tax          : Rs. 4750.00
Net Salary   : Rs. 92750.00
=====

=====
EMPLOYEE PAYSIP
=====
Emp ID      : 1002
Name       : Rahul Patil
Department  : HR
Grade      : A
Status     : Active
-----
Basic Salary : Rs. 60000.00
Allowance    : Rs. 24000.00
Gross Salary : Rs. 84000.00
Tax          : Rs. 3400.00
Net Salary   : Rs. 80600.00
=====

```

Bonus — Struct Padding

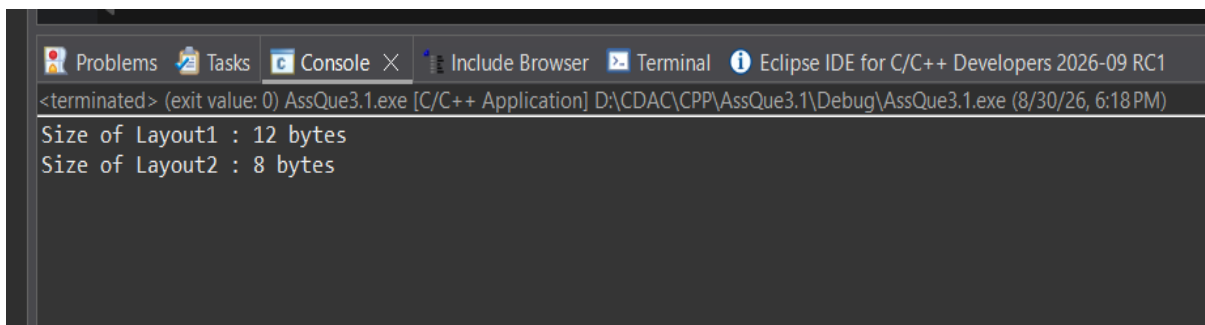
```
//=====
=====
// Name      : 1.cpp
// Author    :
// Version    :
// Copyright  : Your copyright notice
// Description : Hello World in C++, Ansi-style
//=====
=====
```

```
#include <iostream>
using namespace std;
```

```
struct Layout1
{
    char c1;
    int i;
    char c2;
};
```

```
struct Layout2
{
    int i;
    char c1;
    char c2;
};
```

```
int main()
{
    cout << "Size of Layout1 : " << sizeof(Layout1) << " bytes" << endl;
    cout << "Size of Layout2 : " << sizeof(Layout2) << " bytes" << endl;
    return 0;
}
```



The screenshot shows the Eclipse IDE interface with the 'Console' tab selected. The output of the program is displayed in the console window, showing the size of the two struct layouts. The first line of output is 'Size of Layout1 : 12 bytes' and the second line is 'Size of Layout2 : 8 bytes'. The console title bar indicates the application is 'AssQue3.1.exe' running in debug mode.

```
<terminated> (exit value: 0) AssQue3.1.exe [C/C++ Application] D:\CDAC\CPP\AssQue3.1\Debug\AssQue3.1.exe (8/30/26, 6:18 PM)
Size of Layout1 : 12 bytes
Size of Layout2 : 8 bytes
```